

## Exercise Set 6

20 October 2025

# 1 SMT Solving

## 1.1 SAT solving

- Consider the following formula

$$\phi = (p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee \dots \vee (p_n \wedge q_n)$$

How many clauses does the equivalent CNF formula contain?

- Given

$$\phi = (\neg p \vee q \vee r) \wedge (s \vee p \vee r) \wedge (\neg q \vee \neg r) \wedge (\neg p \vee s) \wedge (\neg q \vee r) \wedge (q \vee \neg s \vee r)$$

is  $\phi$  satisfiable?

- Given

$$\phi = r \wedge (q \vee \neg r) \wedge (s \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s)$$

show that  $\phi$  is unsatisfiable.

**Candies:** The candy box is empty!

There are three children in the room (Alice, Bob and Carole). Here some reasoning to track who is the guilty candies eater.

1. One of the three children ate some Candies
2. If Alice has eaten some candies, then so has Bob
3. If Alice did not eat candies, then neither did Bob
4. if Alice and Bob ate candies, then so did Carole
5. If neither Alice nor Bob have eaten candies, then Carole would not have eaten candies

Therefore, the three of them have eaten candies.

Model the problem as a propositional logic formula. Is the reasoning correct?

**Neg and Pos:** we say that a clause is *positive* (respectively *negative*) if and only if it contains only positive literals (no  $\neg$  in the clause. Respectively for *negative* they all appear with  $\neg$ ). Let  $E$  be a set of unsatisfiable clauses without the empty clause. Show that  $E$  contains at least a positive clause and a negative clause.